

AFCI-Bench - results delivery

Compiled 2026-09-26 from study-results/ on branch study-v2, pinned to evidence commit 7822e41b18fd (*study(backstage): halt task qualification V2 after 1 of 15 observations*) - the last change to the evidence this package reports.

Reporting and export only. No benchmark observation was executed to produce this package, no model was invoked, and no task definition, architecture document, scorer, threshold, run plan, condition, raw run artifact or prior analysis was changed. Every figure below was recomputed from the 181 run/attempt rows and then checked against the frozen per-experiment analysis artifacts:

171 checks, 0 mismatches. The full check list is in the workbook sheet 20_RECOMPUTATION_AUDIT.

Claims are tagged **[FACT]** (a measured value, reproducible from the artifacts),

[INTERPRETATION] (a reading of those facts) and **[LIMITATION]** (a bound on what the facts can support).

1. Study objective

AFCI asks a narrow question. If you hand a coding agent an explicit, machine-readable description of a repository's architecture - the MAD - does it build better, or more cheaply, than an agent given the task alone?

The programme operationalises that as two conditions held identical in every other respect:

- **C1** - the task, and nothing else.
- **C4** - the same task, plus the explicit Machine-readable Architecture Document.

Two constructs are measured, and they are never combined into one score:

- **architectural conformance** - does the run violate the architecture

opportunities its own changes made relevant? This is the study's actual construct.

- **efficiency** - tokens, wall time, exploration, tool effort and provider cost.

This is the secondary construct, and it is the one with a clean pilot.

2. Experiments conducted

#	experiment	model	runs	status	decision
1	V1 original study	"Opus 7"	48	COMPLETE, PUBLISHED, superseded	published, then superseded by v1 limitations L1-L7
2	PT08 C1 difficulty diagnostic	claude-sonnet-5	3	COMPLETE	PT08 = REVISE; benchmark investment = CONTINUE
3	PT09 instrument qualification	claude-sonnet-5	3 (+1 infra-invalid)	COMPLETE	FAIL / ARCHITECTURE FLOOR -> STOP / REASSESS
4	PT10 instrument qualification	claude-sonnet-5	3	COMPLETE	REVISE / WEAK PRESSURE -> STOP / REASSESS
5	Efficiency pilot, Attempt 1	claude-sonnet-5	9 attempted of 36	ABORTED	excluded wholesale; no analysis was ever performed
6	Efficiency pilot, Attempt 2	claude-sonnet-5	36	COMPLETE	STOP - NO EFFICIENCY SIGNAL JUSTIFIES FULL-SUITE EXPANSION
7	Lower-capability model pilot	claude-haiku-4-5-20251001	18	COMPLETE	NO LOWER-MODEL SIGNAL - DO NOT EXPAND THE SYNTHETIC LOWER-MODEL MATRIX
8	Backstage real-repository pilot, Attempt 1	claude-sonnet-5	7 attempted of 18	HALTED	excluded wholesale; no analysis was ever performed
9	Backstage real-repository pilot, Attempt 2	claude-sonnet-5	18	COMPLETE	NO ARCHITECTURE SIGNAL - DO NOT AUTOMATICALLY EXPAND
10	Backstage task qualification V1 (<i>instrument qualification / C1-only / pre-treatment / not AFCI treatment evidence</i>)	claude-sonnet-5	15 C1 (+1 pre-delivery infra-invalid)	COMPLETE	INSUFFICIENT QUALIFIED TASKS - 1 of 5 qualified
11	Open-source complexity study	TBD	0	NOT STARTED	none - no runs exist and no result is pre-populated

All three completed pilots were judged by a decision rule frozen **before** the data they judge existed, and all three rules returned a negative verdict. None was changed afterwards. The task qualification (row 10) is not a pilot and judges no treatment: it applied a frozen **instrument** rule to C1 runs only, to decide which tasks a future study may use.

3. Evidence and run inventory

[**FACT**] Mechanically counted from AFCI_MASTER_RUN_RESULTS.csv.

measure	count
total run/attempt records	181
distinct run identities	178
completed observations	159
functionally valid observations	93
explicitly functionally invalid	23
no functional verdict captured at all	65
diagnostic / qualification observations	45
infrastructure-invalid or damaged records	9
refused	1
excluded from every analysis	125
eligible for any analysis	56
confirmatory observations	0

The three "missing" run identities are not a bookkeeping error: the PT08 diagnostic minted one run id for all three of its repetitions, and one Attempt-1 collision pair shares a single id. Both are recorded defects, kept as found.

experiment	rows	status COMPLETE	functionally valid	eligible	token coverage	cost coverage	architecture coverage
V1_ORIGINAL	48	48	not captured	0	0/48	0/48	0/48
V2_PT08_DIAGNOSTIC	3	3	3	0	0/3	0/3	3/3
V2_PT09_QUALIFICATION	4	3	3	0	0/4	0/4	3/4
V2_PT10_QUALIFICATION	3	3	3	0	0/3	0/3	3/3
V2_EFF_ATTEMPT1	9	0	not captured	0	0/9	0/9	0/9
V2_EFF_ATTEMPT2	36	35	34	34	35/36	19/36	0/36
V2_LOWER_MODEL_PILOT	18	18	17	16	18/18	18/18	18/18
V2_BACKSTAGE_PILOT	7	0	2	0	7/7	7/7	5/7
V2_BACKSTAGE_PILOT_ATT EMPT_2	18	18	8	6	18/18	18/18	18/18
V2_BACKSTAGE_TASK_QUAL IFICATION_V1	16	15	15	0	15/16	16/16	15/16
V2_BACKSTAGE_TASK_QUAL IFICATION_V2	2	1	0	0	1/2	2/2	1/2
V2_BACKSTAGE_TASK_QUAL IFICATION_V2_R1	17	15	8	0	17/17	17/17	15/17

V2_EFF_ATTEMPT1 shows 0 under *status COMPLETE* because its rows carry INTACT_GOVERNED_OBSERVATION (7 rows) and DAMAGED_* (2 rows) instead - a deliberately distinct status, because those runs were never admitted as observations. The experiment registry counts the same 7 rows as "completed runs"; both readings are shown rather than reconciled away.

[**LIMITATION**] All 56 eligible rows belong to three

non-confirmatory pilots. No p-value, confidence interval, effect size or power estimate exists anywhere in this programme, and none would be defensible from 16, 8 and 3 paired blocks at 3 repetitions.

4. V1 historical results

Design. 12 tasks x {baseline, AFCI} x {non-reset, reset} = 48 runs, one run per cell, model labelled "Opus 7". Published as the ASE 2026 artifact; Zenodo DOI 10.5281/zenodo.19757261.

[FACT]

metric	baseline	AFCI	change	per-task direction
non-reset code churn (mean)	576.58	955.42	+65.7%	AFCI higher in 12/12
non-reset test churn (mean)	172.67	244.00	+41.3%	AFCI higher in 12/12
reset true drift, delta CodeLOC (mean)	307.7	670.5	+117.9%	AFCI lower in 1/12
CI pass	100%	100%	0.0 pp	48/48

[INTERPRETATION] Read at face value, v1 contradicts the hypothesis: AFCI increased churn on every task and was more reset-inconsistent on 11 of 12. It should not be read at face value.

[LIMITATION] Seven recorded limitations, each of which independently undermines inference:

id	limitation	consequence
L1	the tree was never reset between tasks, so edits accumulate across T01-T12	runs are non-independent ; the churn trend is confounded with accumulation
L2	run_one_v1.sh does not invoke a model - it snapshots the working tree	generation and diff capture are not coupled
L3	AFCI-Guard matched literal libs/core paths while the code uses @afci-bench/* aliases	the guard's regexes never fired; its zeros are not measurements
L4	docs/ARCH_RULES.yml is 0 bytes	the machine-checkable architecture rules did not exist
L5	layer_jaccard is a constant 1.0	the metric cannot vary, so it measures nothing
L6	no per-task acceptance oracle beyond npm run ci, which is saturated True	the success gate does not discriminate
L7	npm ci fails on the clean v1 base	deterministic install is broken at the base tag

Also: the v1 baseline ran against a repository whose architecture was visible in the tree, so the "no architecture context" arm was not architecture-free.

v1 is historical / exploratory evidence. No v1 number may be pooled with, or contrasted against, a v2 number.

5. Architecture diagnostic results

These three experiments measure **instruments**, not treatments. A run-purpose firewall enforced by the record schema keeps them out of every confirmatory dataset, treatment-effect estimate and power calculation.

[FACT]

task	runs	functional	applicable opportunities	target violations	classification	consequence
PT08	3	3/3 (15/15 acceptance cases each)	3	0/3	functional ceiling + architecture floor	PT08 = REVISE; benchmark = CONTINUE
PT09	3	3/3 (14/14 each)	3	0/3	FAIL / ARCHITECTURE FLOOR	STOP / REASSESS
PT10	3	3/3 (8/8 each)	3	1/3	REVISE / WEAK PRESSURE	STOP / REASSESS

PT10's single violation is the only target architecture violation recorded anywhere in the programme: one forbidden import, on one line, in one already-existing file, severity blocker, confidence certain, automated. The opportunity id, rule id, forbidden scopes and anchor path are hidden evaluator semantics and stay in the private evaluator repository. **No numeric result is withheld.**

[INTERPRETATION] Three independent instruments at or near the architecture floor on this substrate is evidence about the **substrate**, not only about the tasks. A baseline at the floor has no room beneath it, so no treatment however good can be discriminated.

[LIMITATION] Three observations per instrument. No power calculation justifies that count and none is implied.

6. Sonnet efficiency pilot (Attempt 2)

Design. 3 tasks (PT01, PT04, PT07) x {C1, C4} x {non-reset, reset} x 3 repetitions = 36 runs, claude-sonnet-5, CLI 2.1.229, block-paired with randomised within-block condition order.

[FACT] Execution and validity:

	value
scheduled / executed	36 / 36
completed	35
refused	1 (sequence 12)
functionally valid	34 - C1 17/18 , C4 17/18
paired-eligible blocks	16 of 18 (minimum required 12)
runs entering the paired analysis	32
valid but unpaired	2

[FACT] Endpoints. Every figure is the median of per-block C4/C1 ratios; 1.0000 is parity and above 1.0000 means C4 cost more.

endpoint	n	median C4/C1	C4 lower	PT01	PT04	PT07
TOTAL_INPUT_TOKENS (primary)	16	1.4014	4/16	1.3372	0.9325	1.5419
MODEL_WALL_SECONDS	16	1.2660	4/16	1.2815	0.9834	1.7018
EXPLORATION_CALLS	16	1.3939	3/16	1.0714	1.3333	1.5130
TOTAL_TOOL_CALLS	16	1.1864	5/16	1.1000	0.9200	1.3867
UNIQUE_FILES_READ	16	1.2111	0/16	1.1429	1.3333	1.3571
EDIT_AND_WRITE_CALLS	16	1.9167	0/16	1.6000	1.3333	2.0000
CI_COMMAND_RUNS	16	1.0000	1/16	1.0000	1.0000	1.0000
TOTAL_OUTPUT_TOKENS (non-reset only)	9	1.4671	1/9	1.4671	1.0423	2.5088
provider cost USD (non-reset only)	9	1.4928	2/9	1.4438	0.9511	1.8851

By arm, on the primary endpoint: **NON_RESET 1.5582** (n=9) and **RESET 1.3776** (n=7). PT04 is the only task where C4 was cheaper on median.

[FACT] One signal ran the other way. Reset overhead is a run's metric under reset divided by the same task and condition without reset; lower is better.

endpoint	tasks where C4's recovery overhead is lower
TOTAL_INPUT_TOKENS	2 of 3 (PT01, PT07)
MODEL_WALL_SECONDS	3 of 3 (PT01, PT04, PT07)
EXPLORATION_CALLS	2 of 3 (PT01, PT07)
TOTAL_TOOL_CALLS	2 of 3 (PT01, PT07)

[FACT] The frozen decision. Both functional guardrails and the minimum-pairs gate passed; every token and secondary-endpoint threshold failed. The RESET-SPECIFIC GO branch failed only on its overall token clause - its two reset-overhead clauses passed.

Outcome (rule 11.4): STOP - NO EFFICIENCY SIGNAL JUSTIFIES FULL-SUITE EXPANSION. `efficiency_claim_made = false`.

[LIMITATION] AFCI_EFFICIENCY_PILOT is a **cost-only** purpose under its own frozen governance. **No architecture endpoint was produced and none may be inferred from it.** LOC churn was likewise not captured.

7. Haiku lower-model pilot

Design. The same three tasks on `claude-haiku-4-5-20251001`, non-reset only, 3 repetitions, 18 runs. The point was to test the most obvious explanation for the Sonnet null - that a strong model already knows what the MAD would tell it. Quality and efficiency are two **independent** channels, combined into no single score.

[FACT] Execution and validity: 18 of 18 executed, 18 COMPLETE, **0 refused**, **0 infrastructure-invalid**. 17 of 18 functionally valid - C1 **8/9**, C4 **9/9**. The single invalid run, PT04/C1/R2, failed all four semantic acceptance cases and strands its valid C4 partner, leaving **8 of 9** paired-eligible blocks.

[FACT] Architecture quality:

arm	runs	applicable opportunities	violated	target-violation runs
C1	9	9	0	0

arm	runs	applicable opportunities	violated	target-violation runs
C4	9	9	0	0

Identical by task: 3 applicable, 0 violated in both arms for PT01, PT04 and PT07.

[FACT] Efficiency endpoints, 8 paired blocks:

endpoint	n	median C4/C1	C4 lower	PT01	PT04	PT07
TOTAL_INPUT_TOKENS (primary)	8	2.0372	2/8	2.9214	1.7502	0.9038
TOTAL_OUTPUT_TOKENS	8	1.7804	3/8	2.5067	1.4358	0.7171
MODEL_WALL_SECONDS	8	1.5268	3/8	2.3801	1.2307	0.8427
TOTAL_TOOL_CALLS	8	1.5023	2/8	2.0000	1.4174	0.9412
EXPLORATION_CALLS	8	1.7308	1/8	2.0000	2.1683	1.0769
UNIQUE_FILES_READ	8	1.2500	2/8	1.0000	1.0833	1.5000
EDIT_AND_WRITE_CALLS	8	1.1741	2/8	2.0000	1.1741	0.8750
CI_COMMAND_RUNS	8	1.0000	3/8	1.0000	0.7500	0.5000
provider cost USD	8	1.8865	2/8	2.5232	1.5777	0.8078

PT07 is the one task where C4 was directionally cheaper on tokens, time, output and cost. PT01 was the most expensive for C4 on every endpoint.

[FACT] Rework over the 8 paired blocks: edit calls 54 vs 66, files re-edited 16 vs 20, turns used 229 vs 309, files changed in the worktree diff 25 vs 30, net lines +1,621 vs +1,746. Agent-initiated CI command runs went the other way: 13 for C1 against 8 for C4.

[FACT] The frozen decision:

signal	criterion	observed	verdict
QUALITY	12.1.1 C4 has fewer target architecture-violation runs overall	C4 0 vs C1 0	FAIL
QUALITY	12.1.2 improvement in at least 2 of 3 tasks	improved: none	FAIL
QUALITY	12.1.3 functional guardrail	C4 valid 9 vs C1 valid 8; deficit -1 <= 1	PASS
EFFICIENCY	12.2.1 functional guardrail	C4 valid 9 vs C1 valid 8; deficit -1 <= 1	PASS
EFFICIENCY	12.2.2 at least one endpoint threshold met	median TOTAL_INPUT_TOKENS 2.0372 < 1.00; median EXPLORATION_CALLS 1.7308 <= 0.80; median TOTAL_TOOL_CALLS 1.5023 <= 0.85	FAIL

NO LOWER-MODEL SIGNAL - DO NOT EXPAND THE SYNTHETIC LOWER-MODEL MATRIX

[LIMITATION] Criterion 12.1.1 fails on a **tie at zero**, not on C4 being worse. Neither arm violated anything, so there was nothing to discriminate. The rule requires *strictly fewer* violations, so a tie scores FAIL. That is a rule outcome, not a finding about AFCI: the quality channel produced **no information**.

8. Backstage real-repository pilot (Attempt 2)

Design. The first AFCI experiment on a **real, large, ambiguous** open-source repository rather than the synthetic substrate: Backstage at f285f6e4, which predates the 2026-02-15 contamination boundary. 3 tasks (T1, T2, T5) x {C1, C4} x NON_RESET x 3 repetitions = 18 runs, 9 paired blocks, claude-sonnet-5 at effort high, CLI 2.1.229, 96-turn ceiling. Attempt 1 was halted at 7 of 18 and is **excluded wholesale**; attempt 2 is a wholly new execution with new run ids and new destinations.

[FACT] **Execution.** 18/18 executed in the committed order, 18 valid, **0 infrastructure-invalid, 0 retries**, 18 unique sessions, 0 scientific modifications. Model requested and resolved claude-sonnet-5 on every row; effort high read back from two independent channels over 2,479 hook firings; runtime context CLEAN with loaded context empty on all four fields; export tree 4dfadc10... verified per run. Captured provider cost **\$32.19**.

[FACT] **Primary endpoint - architectural placement.** 1 applicable opportunity per run, 18 across the set.

scope	C1 target-violation runs	C4 target-violation runs
OVERALL	2 / 9	1 / 9
T1	0 / 3	0 / 3
T2	0 / 3	0 / 3
T5	2 / 3	1 / 3

[FACT] Functional endpoint.

scope	C1 valid	C4 valid	paired-valid blocks
OVERALL	4 / 9	4 / 9	3 / 9
T1	0 / 3	0 / 3	0 / 3
T2	2 / 3	3 / 3	2 / 3
T5	2 / 3	1 / 3	1 / 3

[FACT] The frozen decision (SL-V2-BACKSTAGE-PILOT-01 S11.1, all three required):

clause	statement	observed	verdict
11.1.1	C4 has FEWER target-violation runs than C1 overall	C1=2; C4=1	PASS
11.1.2	C4 has FEWER target violations in >= 2 of 3 tasks	C4 fewer in 1 of 3 tasks (T5)	FAIL
11.1.3	C4 FUNCTIONAL_VALID no more than 1 below C1	C1=4; C4=4; difference=0	PASS
DECISION	ARCHITECTURE SIGNAL requires all three	NO ARCHITECTURE SIGNAL - DO NOT AUTOMATICALLY EXPAND	NO SIGNAL

NO ARCHITECTURE SIGNAL - DO NOT AUTOMATICALLY EXPAND

[LIMITATION] Criterion 11.1.2 fails on **ties at zero**, not on C4 being worse. T1 and T2 produced zero target violations in *both* arms, so neither can show C4 as "fewer"; only T5 discriminated. Two of three tasks sat at an architecture floor - the same failure mode the Haiku pilot hit, here partial rather than total.

[LIMITATION] T1 is a **task-instrument failure, not a model result**. All six T1 runs - both arms, all three repetitions - passed exactly 3 of 4 semantic cases and failed the **same single case every time**, while both controlled T1 references pass that case in the frozen reference matrix. The oracle is satisfiable; the model-facing task statement does not ask for the behaviour it checks. T1 therefore contributes 0 functionally valid runs *and* 0 target violations to either arm - inert on both endpoints. Nothing was changed in response; the task bytes, oracle, scorer and rule stand exactly as frozen.

[FACT] MAX_TURNS. 2 of 18 reached the 96-turn ceiling, one per arm, both on T5. The 64->96 raise worked: attempt 1 hit its ceiling on 4 of its 5 valid rows. MAX_TURNS is a scientific outcome and was never a retry reason.

[FACT] Secondary efficiency (S11.2, cannot override S11.1), median C4/C1 over functionally valid paired blocks:

endpoint	median C4/C1	coverage	C4 lower
TOKEN_RATIO	0.9672	3/3	2/3
WALL_RATIO	1.0551	3/3	0/3
EXPLORATION_RATIO	0.875	3/3	2/3
TOTAL_TOOL_RATIO	1.0159	3/3	0/3
COST_RATIO	0.9497	3/3	2/3
UNIQUE_FILES_READ_RATIO	0.9231	3/3	2/3
EDIT_WRITE_RATIO	0.9286	3/3	2/3

[LIMITATION] Coverage is **3 of 9** blocks, because no T1 run is functionally valid. At n=3 a single block moves every median. These figures are descriptive only and could not have rescued a failed S11.1 in any case.

[FACT] The network preflight prevented a repeat of the attempt-1 loss. 20 preflight refusals occurred, every one VPN_ADAPTER_UP - the adapter state that severed two attempt-1 observations. Each refused **before the task was delivered**: no artifact directory, no workspace, no provider call, no observation consumed, \$0. The cost was wall-clock only.

9. BACKSTAGE TASK QUALIFICATION V1

Every row in this section is: instrument qualification / C1-only / pre-treatment / not AFCI treatment evidence.

Design. SL-V2-BACKSTAGE-TQ-01, frozen before its first observation. It asks which candidate Backstage tasks a **future** AFCI study may use, and runs the baseline condition **C1 only**: no C4 observation, no architecture packet, no treatment effect. 5 candidates x 3 C1 observations = 15, claude-sonnet-5 at effort high, CLI 2.1.229, NON_RESET, 96 turns, on the same frozen substrate. Backstage Attempts 1 and 2 are unchanged and never pooled with it.

Why it exists. Attempt 2 showed that the instrument, not only the model, limited what could be seen: T1 was functionally underspecified, T2 sat at an architecture floor, and only T5 discriminated.

- **Q1 BTQ-T1R is T1 repaired** under a new identity. All six Attempt-2 T1

runs failed the same semantic case because the statement never said a returned value must be **synchronous**; one sentence now says so, and no architecture guidance was added. T1's original bytes are untouched.

- **Q2 BTQ-T5 is T5**, byte for byte.
- **Q3-Q5** are newly mined, real post-substrate Backstage changes.
- **T2 is RETIRED** (architecture floor in Attempt 2) and was not substituted.

Every candidate passed static qualification before any model run: the untouched substrate fails the functional check and scores 0/0; a legal reference is functional PASS with 1 applicable, 0 violated; a violating reference is functional PASS with 1 applicable, 1 violated; both pass the model-visible gate.

[FACT] The frozen rule - QUALIFIED iff FUNCTIONAL_VALID >= 2 of 3 **and** target violation >= 1 of 3:

slot	task	rule family	FUNCTIONAL_VALID	target violation	MAX_TURNS	status
Q1	BTQ-T1R	extension-point placement / ownership	3 / 3	0 / 3	0 / 3	ARCHITECTURE_FLOOR_REJECT
Q2	BTQ-T5	shared contract / permission ownership	3 / 3	3 / 3	2 / 3	QUALIFIED
Q3	BTQ-C02	shared contract / permission ownership	3 / 3	0 / 3	1 / 3	ARCHITECTURE_FLOOR_REJECT
Q4	BTQ-C04	package / role ownership	3 / 3	0 / 3	0 / 3	ARCHITECTURE_FLOOR_REJECT
Q5	BTQ-C05	shared contract / API ownership	3 / 3	0 / 3	0 / 3	ARCHITECTURE_FLOOR_REJECT

INSUFFICIENT QUALIFIED TASKS - 1 of 5 qualified (BTQ-T5). Fewer than three: the phase stops, with no further mining.

[FACT] Reading a zero. Every final observation had exactly one applicable architecture opportunity, so each 0 means the legal placement was made every time the decision arose - an architecture floor under C1, not an opportunity the instrument missed. The T1 repair worked on the functional axis (3/3 valid against 0/6 for the original T1 in Attempt 2) but exposed no baseline pressure.

[LIMITATION] Selection. This qualification intentionally selects tasks with measurable baseline architecture pressure. A future treatment study built on its selection estimates AFCI behaviour on ARCHITECTURE-PRESSURE-QUALIFIED tasks and must not be presented as an unbiased estimate over arbitrary Backstage development tasks. The tasks are

pressure-qualified by a rule declared in advance, not cherry-picked; with fewer than three qualified, no task was selected.

[FACT] Execution. 15 valid final observations, all functionally valid; 1 infrastructure-invalid attempt; captured provider cost

\$33.72. One attempt failed **before its task was delivered** (host standby during workspace preparation, no model invoked, \$0): deviation SL-V2-BACKSTAGE-TQ-01-D1 re-derived WORKSPACE_PREPARATION_FAILED from the attempt's own artifacts and the cell completed on its pre-authorized attempt-2 identity.

9B. BACKSTAGE TASK QUALIFICATION V2 - HALTED

Every row in this section is: instrument qualification / C1-only / pre-treatment / not AFCI treatment evidence. None is a qualification result.

Design. SL-V2-BACKSTAGE-TQ-02, pre-registered before its first observation: five **new** mined candidates (Q6-Q10), 3 C1 observations each, the same model, effort, CLI version, turn ceiling and substrate as V1, looking for at least two tasks to join BTQ-T5. It applied V1's lesson - its floors followed an obvious nearby precedent - by mining for changes where the nearest code points away from the documented owner. All five candidates passed static qualification before data (reference matrix, visible gate, task/oracle alignment, leakage audit, real preparation). Two of them prepare only under a pre-registered warm-up report restore, which puts back committed API reports that the frozen warm-up rewrites on the untouched substrate.

[FACT] Halted after 1 of 15 (SL-V2-BACKSTAGE-TQ-02-H1). The runtime pin launched the frozen 2.1.229 binary by an extensionless path. The version and digest checks passed, but Claude Code's built-in Grep tool re-spawns its own executable, which Windows cannot resolve without an executable extension, so the tool could not start. V1 ran the same version under its usual executable name and never hit this (93 successful Grep/Glob results, 0 such errors); no pre-data gate exercised a built-in tool.

seq	slot	task	status	delivered	functional	turns	files changed	cost
1	Q6	BTQ2-C01	valid under the frozen classifier; produced under the defect	yes	FAIL (0/5)	34/96	0	\$1.35
2	Q7	BTQ2-C02	stopped before task delivery	no	-	-	-	\$0.00

HALTED - no candidate status determined; 0 tasks newly qualified. The frozen rule's first branch applies: **INSUFFICIENT ADDITIONAL QUALIFIED TASKS**, reached by halt, not by measurement. BTQ-T5 remains the only qualified task.

[FACT] Decision. Offered a fix under a recorded deviation (launch the byte-identical binary under an executable name, prove the tools, classify every observation launched by the extensionless path as infrastructure-invalid, resume on the pre-authorised second attempts), continuing unchanged, or stopping, the Study Lead stopped the phase. The five candidates remain statically qualified instruments; using them needs a new pre-data decision. That decision followed: the clean re-execution V2-R1, section 9C.

[LIMITATION] Selection. Unchanged from V1: qualification selects tasks with measurable baseline architecture pressure, so any study built on it estimates AFCI behaviour on architecture-pressure-qualified Backstage tasks only.

9C. BACKSTAGE TASK QUALIFICATION V2-R1 - CLEAN RE-EXECUTION OF V2

Every row in this section is: instrument qualification / C1-only / pre-treatment / not AFCI treatment evidence.

Why. V2 was excluded wholesale because of a scientific-runtime instrumentation defect (section 9B).

SL-V2-BACKSTAGE-TQ-02-R1, pre-registered before its first observation, re-ran all five frozen V2 candidates from scratch

- the same statements, oracles, scorers, references and rule families - with a

new seed and new identities, after a pre-data runtime correction: the same 2.1.229 bytes, launched as a byte-identical .exe copy. Before any scientific task, V2's static evidence was re-verified and reproduced exactly, real preparation was re-proved, and a real, non-scientific smoke test showed Grep and Glob working under the corrected executable, while the extensionless path reproduced V2's failure.

slot	task	architecture rule family	FUNCTIONAL_VALID	target violations	MAX_TURNS	cost (USD)	status
Q6	BTQ2-C01	family E: extension contract placement / ownership	0 / 3	0 / 3	2 / 3	8.52	BOTH_REJECT
Q7	BTQ2-C02	family P: package / role ownership	1 / 3	0 / 3	3 / 3	13.64	BOTH_REJECT
Q8	BTQ2-C03	family P: package / role ownership	3 / 3	0 / 3	0 / 3	5.36	ARCHITECTURE_FLOOR_REJECT
Q9	BTQ2-C04	family S: shared / public contract ownership	2 / 3	0 / 3	1 / 3	10.67	ARCHITECTURE_FLOOR_REJECT
Q10	BTQ2-C05	family S: shared / public contract ownership	2 / 3	0 / 3	3 / 3	12.71	ARCHITECTURE_FLOOR_REJECT

INSUFFICIENT ADDITIONAL QUALIFIED TASKS - 0 of 5 newly qualified. BTQ-T5 remains the only qualified task, so no three-task set for a future C1/C4 study exists.

[FACT] Reading the zero. 13 of 15 final observations created the one pre-specified architecture opportunity, and every one of them made the legal placement (0 target violations). The other 2 never wrote the task-mandated construct (both functionally invalid), so the placement decision did not arise in them and they cannot hold a violation; the outcome does not depend on them. The zero is an architecture floor under C1 for every candidate, not an opportunity the instrument missed.

[FACT] Deviation SL-V2-BACKSTAGE-TQ-02-R1-D1. One cell (BTQ2-C02, repetition 2) lost both frozen attempts to an expired host sign-in credential before any model turn (0 tokens, \$0). The Study Lead authorised, outcome-blind, one additional identity for that cell only - usable only when both spent attempts' own records prove the authentication failure - plus a credential check before every launch. The cell completed on it; nothing else changed.

[FACT] Integrity. 15 of 15 final observations; 2 infrastructure-invalid attempts (\$0); 167 Grep/Glob calls and 0 built-in tool launch failures; the byte-identical 2.1.229 executable verified before every delivery; model and effort read back on every observation; prior phases, the halted V2 included, verified unchanged after observation 15.

[FACT] Cost. V2-R1 \$51.01 (observations \$50.90, pre-data smoke test \$0.11); the halted V2's \$1.3543 is reported separately; cumulative task qualification \$86.09.

[LIMITATION] Selection. Unchanged: qualification selects tasks with measurable baseline architecture pressure, so any study built on it estimates AFCI behaviour on architecture-pressure-qualified Backstage tasks only.

10. Cost and token findings

[FACT] Token audit. $\text{TOTAL_INPUT_TOKENS} = \text{input_tokens} + \text{cache_creation_input_tokens} + \text{cache_read_input_tokens}$ on every row that carries tokens - checked on all 111 such rows, 0 failures. The MAD's own tokens and all cache traffic stay inside the number.

scope	paired blocks	C1 total input tokens	C4 total input tokens	C1 median/run	C4 median/run	median C4/C1
Sonnet overall	16	10,988,803	14,859,860	698,639	1,040,459	1.4014
Sonnet NON_RESET	9	5,912,725	8,339,159	664,777	1,062,096	1.5582
Sonnet RESET	7	5,076,078	6,520,701	733,633	1,028,101	1.3776
Haiku NON_RESET	8	5,599,750	9,504,643	652,919	1,271,816	2.0372

Token coverage: 111 of 181 run rows carry input-token evidence, 95 carry output tokens. v1, the aborted Attempt 1 and the three diagnostics predate or do not use the token instrumentation; their cells are blank, never 0. The one Backstage task-qualification attempt that failed before delivery invoked no model and has no token record either.

[FACT] Cost audit.

scope	runs	runs with complete cost	captured total	verdict
Sonnet Attempt 2 - all rows	36	19	\$11.1919	PARTIAL COST CAPTURE
Sonnet Attempt 2 - NON_RESET	18	18	\$10.9463	COMPLETE COST COVERAGE
Sonnet Attempt 2 - RESET	17	1	\$0.2456	PARTIAL COST CAPTURE
Haiku lower-model pilot - all rows (NON_RESET by design)	18	18	\$3.8007	COMPLETE COST COVERAGE

The Sonnet RESET row counts 17, not 18: the refused run (sequence 12) carries no reset state at all, so it sits outside both arms. $18 + 17 + 1 = 36$.

paired cost ratio	n	C1 total	C4 total	median C4/C1
Sonnet NON_RESET	9	\$4.5288	\$6.4175	1.4928
Sonnet RESET	0	n/a	n/a	no cost comparison exists
Haiku NON_RESET	8	\$1.3862	\$2.0068	1.8865

Total captured provider cost across all runs carrying cost evidence: \$144.09 over 97 runs.

CAPTURED COST, NOT NECESSARILY TOTAL STUDY COST. 84 of 181 run rows carry no provider-cost record at all: all 48 v1 rows, all 9 Attempt-1 rows, all 10 diagnostic rows, 16 of 17 Sonnet RESET rows, and the one refused run. Money was spent on those runs; the runtime never reported it in a form the record could carry.

[LIMITATION] A reset run's phase A is deliberately interrupted at the checkpoint, **before** its terminal result event - the only place the runtime reports output tokens and cost. For a reset run, input tokens remain **exact** (reconstructed from the streamed assistant messages) while output tokens and cost are **withheld, not zero and not estimated**. The one Sonnet reset run that does carry a cost is PT04/C1/RESET/R1, and it carries one precisely because it never reached its checkpoint - it is functionally invalid and enters no paired figure. There are therefore **zero usable Sonnet RESET cost pairs**, and the 1.4928 figure describes the non-reset arm only.

11. Functional and architecture quality

[FACT]

experiment	C1 valid	C4 valid	paired-valid blocks	semantic cases passed (C1 / C4)
Sonnet Attempt 2	17/18	17/18	16 of 18	68 / 68
Haiku lower-model pilot	8/9	9/9	8 of 9	32 / 36

[FACT] Architecture, over every run that produced a measurement:

experiment	arm	runs	applicable	violated	target-violation runs
PT08 diagnostic	C1	3	3	0	0
PT09 qualification	C1	3	3	0	0

experiment	arm	runs	applicable	violated	target-violation runs
PT10 qualification	C1	3	3	1	1
Haiku pilot	C1	9	9	0	0
Haiku pilot	C4	9	9	0	0
Sonnet Attempt 2	both	36	NOT MEASURED	NOT MEASURED	NOT MEASURED
V1 original	both	48	INVALID FOR INFERENCE	INVALID FOR INFERENCE	INVALID FOR INFERENCE

[INTERPRETATION] Functional acceptance is saturated and architecture sits at a floor. Six instrument/model combinations now sit at or near that floor on this substrate: PT08, PT09 and PT10 for an unguided Sonnet baseline, and PT01, PT04 and PT07 for Haiku in **both** arms.

[LIMITATION] The evidence base is lopsided. The study's *actual* construct - architectural conformance - has been measured only by instruments that all sat at or near the floor, while its *secondary* construct - cost - has one clean pilot. The cost result must not be allowed to stand in for an architecture result.

12. Cross-model interpretation

[FACT] Both sides below are NON_RESET, so the comparison is like-for-like.

metric	Sonnet C4/C1 (n=9)	Haiku C4/C1 (n=8)	difference	reading
TOTAL_INPUT_TOKENS	1.5582	2.0372	+0.4791	Haiku ratio higher
TOTAL_OUTPUT_TOKENS	1.4671	1.7804	+0.3133	Haiku ratio higher
MODEL_WALL_SECONDS	1.4968	1.5268	+0.0300	Haiku ratio higher
EXPLORATION_CALLS	1.5714	1.7308	+0.1593	Haiku ratio higher
TOTAL_TOOL_CALLS	1.3333	1.5023	+0.1690	Haiku ratio higher
UNIQUE_FILES_READ	1.4000	1.2500	-0.1500	Haiku ratio lower
EDIT_AND_WRITE_CALLS	2.0000	1.1741	-0.8259	Haiku ratio lower
PROVIDER_COST_USD	1.4928	1.8865	+0.3937	Haiku ratio higher

channel	Sonnet	Haiku	comparable?
architecture target violations	NOT MEASURED (cost-only purpose)	0/9 in C1 and 0/9 in C4	no
functional validity	17/18 and 17/18	8/9 and 9/9	descriptively only

This is a descriptive comparison between two separate experiments, not a randomised cross-model causal comparison. The two pilots ran under separate purposes, separate schedules and separate frozen analyses and are never pooled. No interaction was estimated, no test was performed, and no pooled model exists.

[INTERPRETATION] The moderator hypothesis predicted that a weaker coding model, having less ability to infer architecture from the repository alone, would find the explicit MAD *more* useful. The prediction did not come true on the primary endpoint: C4's median input-token ratio **rose** from 1.5582 to 2.0372, a difference of 0.4791 in the direction opposite to the prediction, and the quality channel produced nothing at all because both arms sat at zero.

Two of the eight metrics did move the predicted way - UNIQUE_FILES_READ, EDIT_AND_WRITE_CALLS - and both are reported above rather than dropped. Neither is the primary endpoint, both remain **above** 1.0000 on the Haiku side (so C4 still cost more in absolute terms on both models), and with 8 and 9 paired blocks and no test performed, a two-of-eight split is not a result. It is recorded so that the six-of-eight direction is not read as unanimity.

13. What the evidence supports

- [FACT]** On this substrate with `claude-sonnet-5`, explicit MAD injection costs more than it saves across every captured cost dimension except CI invocations.
- [FACT]** It does so without buying functional quality: 17/18 versus 17/18.
- [FACT]** Weakening the model did not reduce C4's relative cost; it raised it, and produced no quality gain because both arms sat at zero violations.
- [FACT]** The reset-recovery direction is the one consistent counter-signal, strongest in wall time (3 of 3 tasks).
- [INTERPRETATION]** The **strong-model ceiling** explanation for the repeated null is **not supported**. Instrument discrimination, not model capability, is the binding constraint on the programme.

6. **[INTERPRETATION]** Expanding either pilot, unchanged, is not justified. Both frozen rules returned a negative verdict and both were honoured.
7. **[INTERPRETATION]** The benchmark machinery works. Sterile execution, model pinning and readback, context auditing, hidden acceptance, the architecture oracle, the artifact firewall, run-identity derivation and the frozen analysis all executed end to end - and caught their own defects twice, in ways that cost real money and were recorded rather than smoothed over.

14. What the evidence does NOT support

1. **We cannot conclude AFCI does not work.** We can conclude it did not reduce cost here. Efficiency is one construct; architectural conformance is the study's actual construct, and the cost pilot **did not measure it at all**.
2. **We cannot attach any statistical confidence to anything.** No p-value, no confidence interval, no effect size, no power estimate.
3. **We cannot generalise past this substrate** - one synthetic 49-file monorepo whose architecture is legible from its own code.
4. **We cannot generalise past the two models used.** `primary_model` is still null; no model has been selected for the study.
5. **We cannot use the reset-recovery signal as a finding.** It is descriptive, from 3 tasks, and it failed the branch that would have made it a GO.
6. **We cannot read anything as a v1 hypothesis test.** L1-L7 make the v1 comparison uninterpretable as a treatment contrast.
7. **We cannot claim any architecture result from the Sonnet pilot.** It produced none.
8. **We cannot claim an architecture result from the Haiku pilot either.** It produced an endpoint, but at the floor in both arms. A tie at zero discriminates nothing and must not be read as AFCI failing to improve architecture.
9. **We cannot claim that model capability moderates AFCI's value.** Two separate experiments compared descriptively, never pooled.

15. Limitations

#	limitation
1	No confirmatory evidence exists. Collection has not begun. The suite-wide protocol is PRE-FREEZE, gate G1 is not passed, and TD-B32, TD-B34, TD-B03 and TD-B19 remain open.
2	No statistical inference of any kind. Every reported figure is a median of paired ratios - a descriptive statistic. No power calculation has ever been run.
3	One synthetic substrate , 49 files, whose layering is inferable from its own import graph and path aliases. A model can often deduce the intended architecture <i>without</i> the MAD, which directly attacks the MAD's marginal value. This is the primary external-validity threat to every v2 finding.
4	No study model has been selected. Both models used so far are provisional.
5	Architecture is measured only where it sat at a floor. The evidence base is lopsided toward cost.
6	Reset cost and output tokens do not exist for reset runs - withheld by construction, not missing by accident.
7	This package is a secondary artifact. Where it and a primary artifact disagree, the primary artifact wins.
8	Task selection by baseline pressure. Backstage task qualification V1 selects tasks with measurable C1 architecture pressure; any study built on it estimates AFCI behaviour on architecture-pressure-qualified tasks and must not be presented as an unbiased estimate over arbitrary Backstage tasks. It qualified 1 of 5 candidates, so no selection exists yet.

16. Current research direction

The chain is short and each link is recorded:

1. **v1 measured the wrong things well enough to notice.** Churn is a proxy, the guard did not fire, the oracle did not discriminate, and runs were not independent. v2 replaced proxies with a hidden functional acceptance oracle and an out-of-band architecture oracle.
2. **PT08 showed the instruments, not the theory, were the bottleneck.** A baseline at the architecture floor cannot discriminate however good the treatment is.
3. **PT09 and PT10 showed it was not specific to PT08.** Two purpose-built replacements landed in the same place.

4. **The efficiency pilot asked a cheaper question that could be answered now** - does the MAD at least pay for itself? - and answered **no**, under a rule frozen before the data existed.
5. **The lower-model pilot weakened the model, and the null held.** A weaker model produced no quality signal and a *worse* cost ratio.
6. **[INTERPRETATION]** The common factor across steps 2-5 is not model strength - that has now been varied and did not matter. What remains is that **a legible synthetic repository gives an agent little to gain from being told its architecture**, whether the agent is strong or weak.

[INTERPRETATION] Repository architectural complexity / ambiguity / context-recovery burden is therefore the next scientifically distinct moderator to investigate. **This is the next hypothesis to test, not a proven one.** It stands by elimination rather than by evidence, and nothing in this package tests it.

[FACT] The first real-repository evidence points the same way. Backstage Attempt 2 returned NO ARCHITECTURE SIGNAL with two of three tasks at an architecture floor, and Backstage task qualification V1 (C1 only, no treatment) then found the floor again on 4 of 5 candidates: only T5 carries measurable baseline pressure. **[INTERPRETATION]** On a large, ambiguous real repository too, `claude-sonnet-5` mostly makes the legal placement decision without being told the architecture, so finding tasks where a baseline agent actually violates the architecture is itself the bottleneck for any C1-vs-C4 study.

17. Next study - open-source architectural complexity

Status: NOT STARTED. No runs exist. No results are pre-populated.

Re-run against real / open-source repositories with genuine architectural complexity, where the architecture is *not* inferable from a 49-file import graph. That directly tests the substrate-legibility explanation - the only one of the two rival explanations still standing.

Its motivation was never contingent on the lower-model outcome: repository complexity is an independent moderator and the lower-model pilot measured nothing about it. Removing the ceiling rival raises its priority; it does not authorise it.

Before any run it needs its own decision record, new tasks, new hidden acceptance packages and new architecture rule linkage. Gate G1 is still not passed and the suite is still not frozen.

Sources: study-results/AFCI_MASTER_RUN_RESULTS.csv, AFCI_MASTER_EXPERIMENT_REGISTRY.csv, and the frozen per-experiment analysis artifacts under 01_v1_original_study/ through 06_lower_model_pilot_completed/. Traceability from any figure here to its raw artifact is in AFCI_Professor_Evidence_Map.md and workbook sheet 18_EVIDENCE_MAP.